

Chapter 6, Problem 1RQ

Problem

What charge is on a 5-F capacitor when it is connected across a 120-V source?

- (a) 600 C
- (b) 300 C
- (c) 24 C
- (d) 12 C

Step-by-step solution

Step 1 of 2

The expression for the charge on a capacitor of capacitance C is,

$$Q = CV \text{ (1)}$$

Here, Q is the charge and V is the voltage across capacitor.

Step 2 of 2

Recall equation (1).

$$Q = CV$$

Substitute 5 for C , and 120 for V .

$$\begin{aligned} Q &= 5 \times 120 \\ &= 600 \text{ C} \end{aligned}$$

Therefore, option (a) is correct.